

FROM TERMINAL TO CLUSTER

Shell survival for your first HPC session

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UNDERSTAND → NAVIGATE → RUN → SCALE

WHY THE SHELL?



REMOTE

The machine may be
somewhere else



SHARED

Many people use
the same system



AUTOMATED

Work must be
repeatable



The shell lets you control a computer you cannot see.

TERMINAL ≠ SHELL ≠ MACHINE



TERMINAL

The window



SHELL

Reads commands



MACHINE

Executes them

kevin@lab-pc:~\$

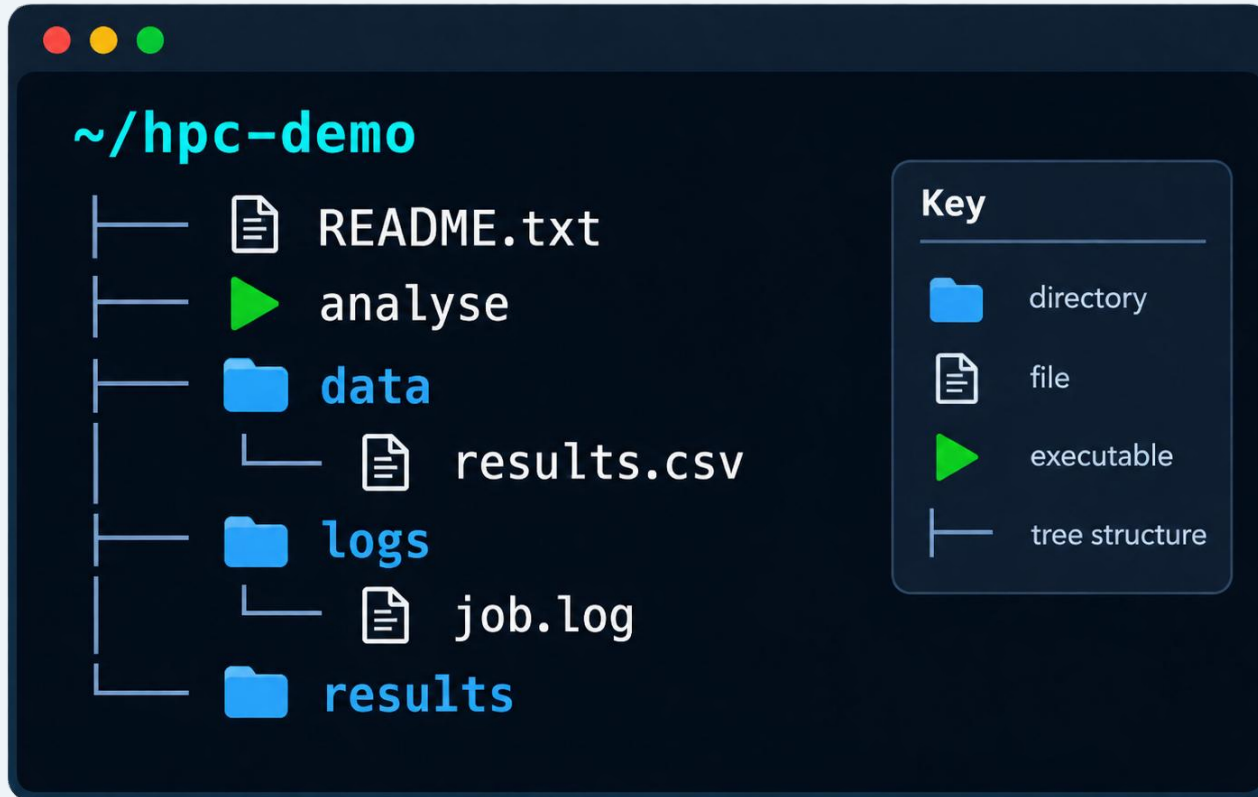
user

machine

location

ready

THE PROJECT TASK



1  **Inspect the input**

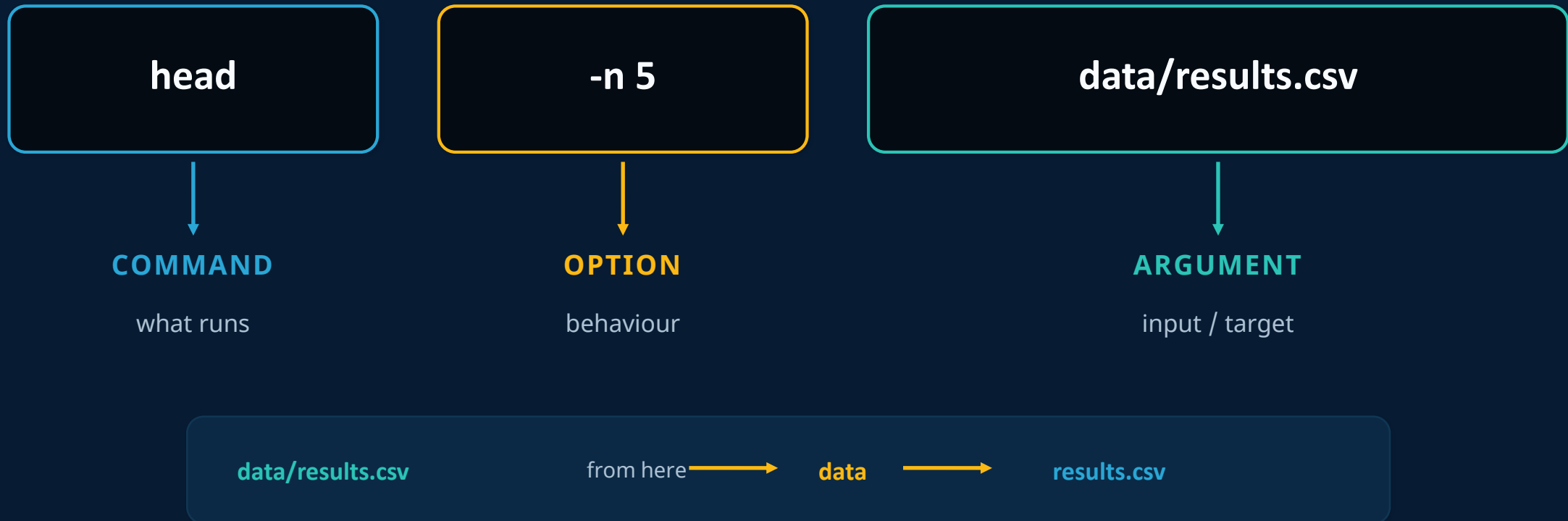
2  **Run the program**

3  **Save the summary**

4  **Count completed runs**

The workload could be a simulation, benchmark, ML job or GPU app.
The interaction pattern is the same.

HOW TO READ A COMMAND



Read the pieces. Do not memorise the whole line.

SAME TERMINAL. SAME SHELL CONCEPTS. DIFFERENT COMPUTER.



Watch the hostname. The window may stay the same; the machine changes.

WHAT CHANGED — AND WHAT DIDN'T?



STAYED THE SAME

- hostname
- pwd
- ls
- cd
- paths
- programs
- input and output



CHANGED

- which machine
- how resources are allocated
- where heavy work runs
- how the workload is launched

The shell is the interface. The scheduler decides where substantial work runs.

SHELL SURVIVAL KIT

ORIENT YOURSELF

**hostname**

Which machine am I on?

**pwd**

Where am I?

**ls**

What is here?

Start here after every remote login.

MOVE & RUN

cd <directory>

Move into a directory

cd ~

Return home

./program

Run the program here

Tab

complete names



previous command

Ctrl+C

stop foreground

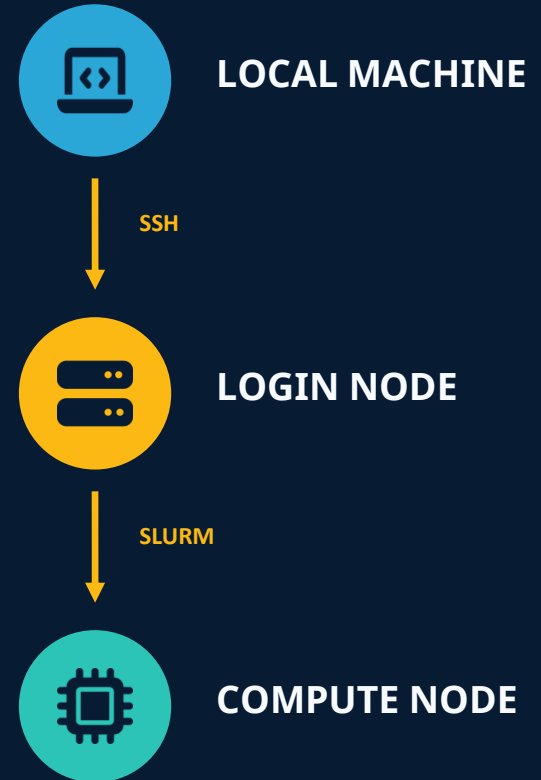
LOCAL → LOGIN → COMPUTE



```
$ hostname  
kevin-laptop
```

```
$ ssh cluster  
$ hostname  
login01
```

```
$ srun -N 1 -n 1 hostname  
compute03
```



Real Parallel Workload

HPL — High-Performance LINPACK



HPL solves a large dense linear-algebra problem.

Classic HPC benchmark. This is a tiny CPU run to show the workflow, not true performance.



```
login01:~/hpl-demo$ srun -n 4 ./xhpl
```

HPL 2.3 - High-Performance Linpack

```
=====
Number of problems      :      1
Problem size (N)       : 12000
Block size (NB)        :   256
Process grid (P x Q)   : 2 x 2
Number of processes    :      4
=====
```

N	NB	P	Q	Time (s)	Gflop/s	Residual	Check
12000	256	2	2	18.74	61.32	2.31e-10	PASSED

```
=====
Finished      : Wed Nov 27 10:42:18 2024
=====
```

```
login01:~/hpl-demo$
```



Performance result

61.32 GFLOP/s



Validation passed

PASSED

Login node → Slurm allocation → Compute node → Parallel workload